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Palaeomagnetism of Precambrian Sedimentary Rocks from NW Scotland and the Apparent Polar Wandering Path of Laurentia

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Summary

Red sedimentary rocks of the Stoer Group (991 My BP) were resampled through their stratigraphic range (600 m at Stoer Bay). The mean direction of magnetization after thermal cleaning at 500° C is 313, +29 ($\alpha_{95} = 8°$) with a corresponding palaeomagnetic pole at 35° N, 234° E. This result is not significantly different from that obtained previously from studies of natural remanent magnetization. A small collection from the base of the overlying Torridon Group (796 My BP) confirms the general reliability of the earlier work on these rocks also, and shows that the large change between the Stoer and Torridon Groups coincides with the unconformity between them. The magnetization of these sediments originated at the time they were laid down. Since NW Scotland used to be part of Laurentia, the Stoer and Torridon Group poles can be compared with those from the late Precambrian of the Canadian Shield. In order to quantify these comparisons, a system of grading poles on a seven-point scale is introduced, and a generalized polar path for the Canadian Shield drawn. The poles from NW Scotland do not fit onto this path, indicating either that there has been slight rotation of NW Scotland as a marginal block of Laurentia, or that a hitherto unsuspected polar excursion occurred immediately following the Grenvillian orogeny.

1. Introduction

Palaeomagnetic study of the Precambrian sedimentary rocks of NW Scotland was initiated by S. K. Runcorn in June 1951, and was the first such study of Precambrian rocks. The present work on these beds has three objectives. The first is to refine the early observations of natural remanent magnetization (nrm) made on 'lower Torridonian' rocks (Irving & Runcorn 1957; Irving 1957) using demagnetization techniques. The second is to determine if the change in direction of magnetization previously used to define the boundary between 'lower' and 'upper' Torridonian (Irving & Runcorn 1957) occurs precisely at the angular unconformity now known (Stewart 1966, 1969) to separate the red sandstones of the Stoer Group (which yielded the original 'lower Torridonian' pole) from those of younger, but still Precambrian Torridon Group (which yielded the original 'upper Torridonian' pole). The third aim is to relate these results to the apparent polar wandering path of the Canadian Shield with which NW Scotland was once continuous.

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2. Geology and sampling

The distribution of rocks and sampling sites are shown in Fig. 1, and stratigraphic sections in Figs 2 and 3. The Stoer Group consists of at least 2300 m of red sandstones and shales, generally dipping 20–25 °W. It contains the Stac Fada Member, which is a volcanic mudflow breccia and the only index horizon in the Group. The Stoer Group rests unconformably on gneisses of the Lewisian basement complex. The Torridon Group consists of at least 7000 m of red sandstones and shales generally dipping at low angles to the east. It is divided into three formations (Diabaig, Applecross, Aultbea) and is overlain by the Lower Cambrian (Peach *et al.* 1907). Both Groups are unmetamorphosed.

Accounts of the geochronology have been given by Moorbath *et al.* (1967) and Moorbath (1969). The Lewisian basement last suffered regional metamorphism at

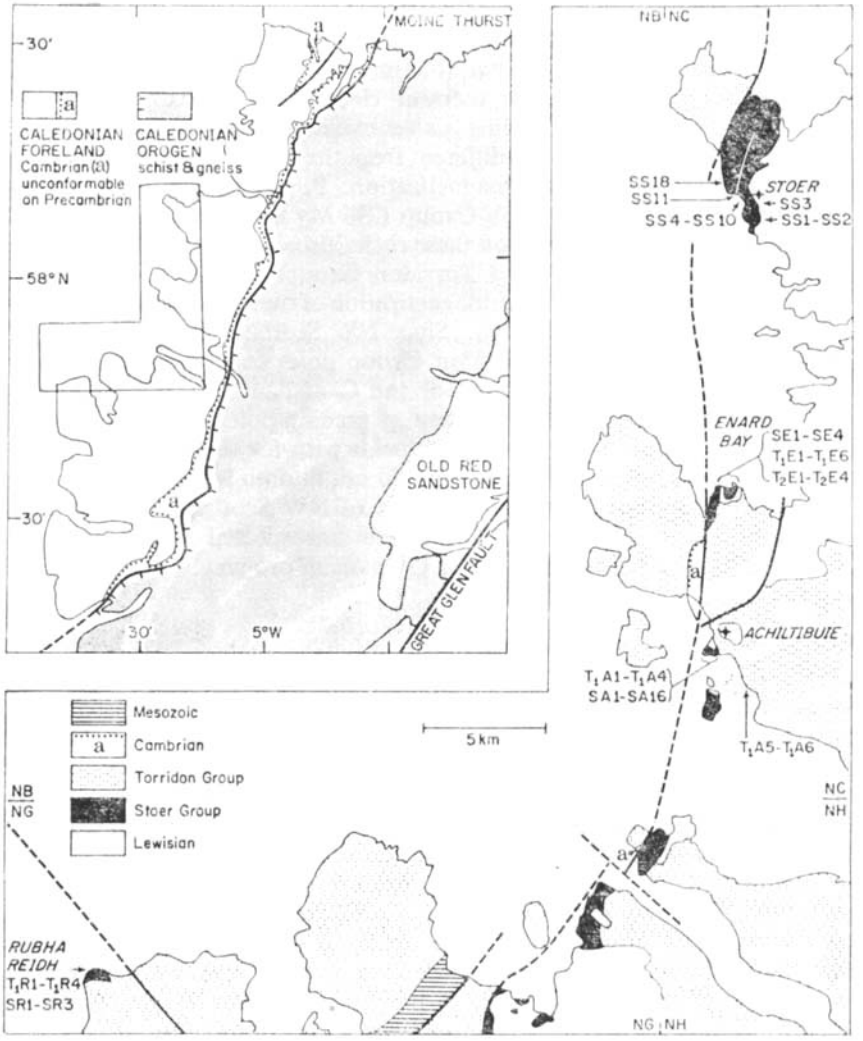


FIG. 1. Sampling sites and geology. Inset shows the relationship of the area to the Caledonian orogen. Sites SS12–17 and SSZ lie between SS11 and SS18 in the order given in Table 1. Boulders in conglomerates at the base of the Torridon Group were sampled at Enard Bay, Rubha Dunain (Achiltibuie) and Rubha Reidh.

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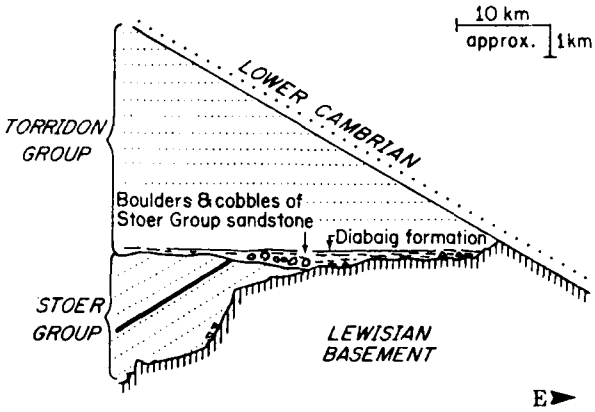


FIG. 2. Generalized section through the Caledonian foreland showing relationships of the Stoer and Torridon Groups to the basement. The heavy line is the Stac Fada Member.

1500–1800 My. Mica and whole-rock K–Ar ages (8 determinations) of 1120 ± 40 My have been reported from parts of the basement, and also from detritus in the Stoer and Torridon rocks. The sediments can not be older than this. Rb/Sr whole-rock measurements on shales and siltstones from the Stoer and Torridon Groups yield excellent isochrons and ages of 991 ± 24 My and 796 ± 24 My respectively (Rb^{87} decay constant $1{\cdot}39 \times 10^{-11}\ yr^{-1}$).

Samples were obtained from beds spaced approximately every 30 m from the base of the Stoer Group to within 0·75 m of the unconformity with the Torridon Group. The lowest 100 m of the Torridon Group was also sampled. From each bed, two to three hand samples were collected, the thickness spanned being 5–100 cm. The time represented by each sandstone site is probably less than 10 yr, but for shales (2 sites only) might reach 1000 yr. Two cores were cut from each sample and one specimen (2·54-cm diameter, 2·2-cm height) from each core. Site directions were obtained by averaging results for cores giving each unit weight.

Four localities were studied; all except Enard Bay had been sampled previously in less detail (Irving & Runcorn 1957). The first locality (19 sites through 600 m) is the north shore of Stoer Bay (Fig. 3). The Stac Fada Member was sampled at SS11.

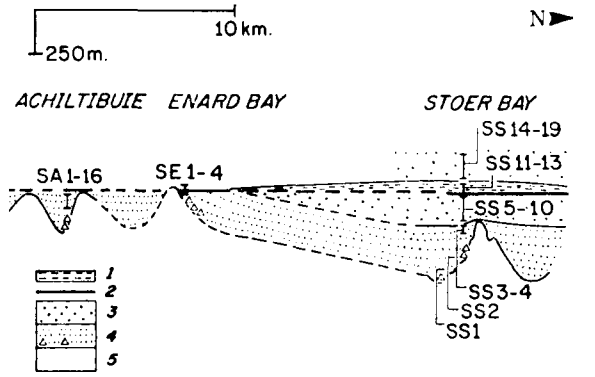


FIG. 3. Stratigraphy of Stoer Group and the relative positions of the sampled sections. Rubha Reidh is not included because the Stac Fada Member does not occur there and precise correlation is impossible. 1 red mudstone and subordinate fine-grained sandstone. 2 Stac Fada Member. 3 Exotic facies, fine to medium grained red sandstone. 4 Marginal facies, locally derived red muddy sandstone and gneiss breccia. 5 Lewisian basement.

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The second locality is at Enard Bay, where the Stoer Group, including the Stac Fada Member were sampled, together with the Diabaig (grey shales and siltstones sites T_1E2 to T_1E6 spanning 5 m) and Applecross (red sandstone sites T_2E1 and T_2E2 from the lowermost 10 m) formations. At the third locality (Rubha Dunain near Achiltibuie, SA1–16) 16 sites were regularly spaced through 85 m of fine to medium sandstone in the Stoer Group. The lowest site (SA1) is about 0·75 m beneath the unconformity with the overlying Torridon Group. The uppermost site (SA16) is at about the same stratigraphic level as SS3 and SS4 at Stoer Bay. At Rubha Dunain the Diabaig Formation was sampled over a 5 m stratigraphic range (T_1A2–T_1A4). Site T_1A3 is about 1 m above the unconformity. Diabaig sediments were also sampled about 1·5 km to the south-east (Fig. 1). The fourth locality is Rubha Reidh, where the Stac Fada Member is absent. Precise correlation with the Stoer Group elsewhere is difficult. Probably SR1–SR3 equate with level SS5 or higher at Stoer Bay. SR1 and SR2 are 8 m apart stratigraphically, and SR3 is about 1 m below the unconformity with the Diabaig Formation. The Diabaig was sampled an unknown distance above its base at two sites (T_1R3–T_1R4) about 4·5 m apart.

Boulders of Stoer Group sandstones occur at the base of the Torridon Group. These were sampled at Enard Bay, Rubha Dunain, and Rubha Reidh, in order to determine the minimum age of the remanent magnetization of the Stoer Group using the conglomerate test. (David 1904; Graham 1949).

3. Stability of nrm and opaque mineralogy of sandstones

An astatic magnetometer (Roy 1963) was used for measuring the remanent magnetization, a muffle-type (Irving *et al.* 1961) electrical furnace constructed by Robertson (1964) for thermal demagnetization, and a solenoid with 3-axis tumbler for alternating field (af) demagnetization (Roy 1973).

Red sandstones of the Stoer and Torridon Groups are generally very stable. Af demagnetization in 1500 Oe (peak) changes the intensity only by small amounts (Fig. 4). Blocking temperatures commonly exceed 600 °C (Fig. 5), although a few thermal decay curves fall more rapidly (SS1, T_1A6) with blocking temperatures as low as 300 °C. Blocking temperatures between 550 and 650 °C have also been observed in red sediments from high in the Torridon Group (Irving 1964, p. 98). These characteristics suggest that haematite is the mineral responsible for the stable remanence.

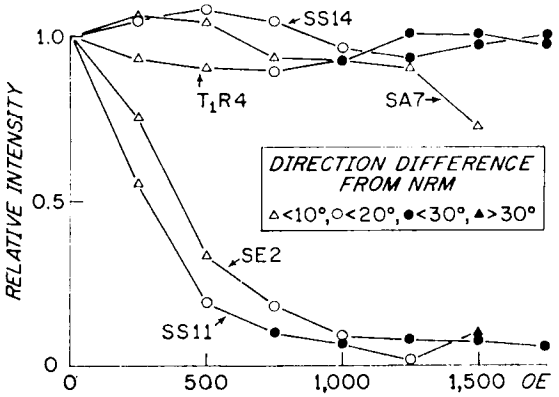


FIG. 4. Alternating field demagnetization. Normalized intensity versus alternating (peak) field in Oersted. SE2 is from the main part of the Stac Fada Member at Enard Bay, and SS11 from the reworked top of the member at Stoer Bay. SS14 and SA7 are from the Stoer Group at Stoer Bay and Achultibuie respectively, and T_1R4 from a natural specular haematite concentrate in the Diabaig Formation.

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Study of polished sections shows opaque minerals in three forms. Firstly, there is specular haematite, which comprises up to 1·5 per cent of the rock by volume, and has a modal grain size of about 100μ. X-ray powder photographs of hand-picked concentrates from the Torridon Group have been identified as αFe_2O_3 (Irving 1957). Moreover, haematite is the only opaque mineral yet identified in the abundant siliceous pebbles of the Torridon Group (Irving 1957; Williams 1969). The specular

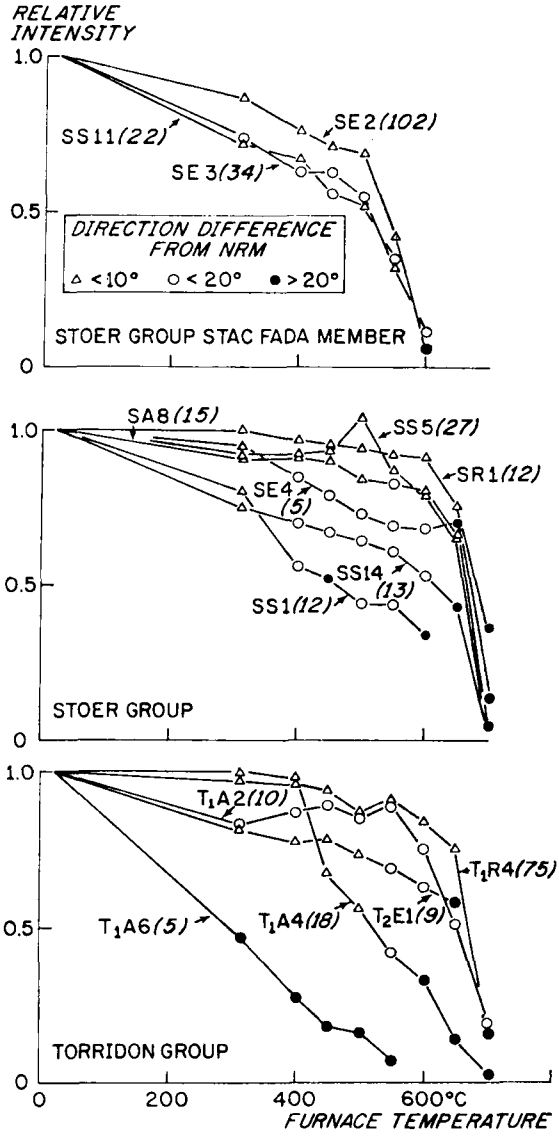


FIG. 5. Thermal demagnetization curves. Specimens were heated to a given temperature, cooled in ' zero ' field (about 20γ), and their magnetization measured. They were then heated to progressively higher temperatures. The intensity, normalized to unity at room temperature, is plotted as a function of furnace temperature. The two lower curves for the Stac Fada Member refer to the reworked top of the unit, and the upper curve to its main part at Enard Bay. In the lower diagram T_1R4, (naturally occurring specular haematite concentrate) T_1A2, T_1A4, and T_1A6 are from the Diabaig Formation, and T_2E1 from the Applecross. The intensity of nrm (μG) is given in brackets.

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haematite is usually disseminated randomly through the rock as detrital grains. It also occurs rarely as films along bedding planes, and around quartz or felspar grains. The detrital specular haematite almost always shows martitization textures—irregular lamellae with widths of 10μ or less forming three-directional patterns. In some sections about 10 per cent of the specular haematite is replaced by ilmenite, and very occasionally grains predominantly composed of ilmenite occur. Partial alteration of ilmenite to rutile is common. The ilmenite usually occurs as needles or bands up to 10μ broad, and occasionally as myrmekitic intergrowths within the host haematite. This latter observation suggests that it was originally intergrown with magnetite, now completely replaced by haematite (Ramdohr 1969, p. 963). In a few grains, the ilmenite lamellae form triangular patterns, suggesting that they were exsolved from an original cubic host, presumably titanomagnetite, now replaced by haematite (*cf.* Phillips in Irving 1957). On the other hand 'oleander-leaf' patterns of ilmenite exsolution lamellae in haematite have been detected by Phillips (in Nairn 1964) who claims they signify unaltered haematite grains derived from igneous rocks, presumably because such patterns are characteristic of a non-cubic host (Ramdohr 1969, p. 176).

Magnetite has not been detected optically, nor is there any indication of it in X-ray photographs (Irving 1957). It has, however, been detected thermomagnetically by Parry (1957) in specular haematite concentrates from the Torridon Group, and by Schwarz (1969) in black banded specimens from the Diabaig Formation (site T_1R4). Schwarz found none in a sample from the Stoer Group (SA13). Whether these results truly indicate the presence of magnetite, or whether they are due to the production of magnetite during a laboratory heating is not certain. In any case, the thermal and af decay curves show that most sandstone has blocking temperatures and coercivities higher than those expected for magnetite, which cannot therefore be a main carrier of the nrm.

The second opaque mineral is goethite, apparently a replacement of specular haematite. Most of the grains that appear black in transmitted light seem to be goethite. Goethite is antiferromagnetic, with a Neel temperature of 120 °C, and can possess a remanence at lower temperatures. It inverts to haematite at about 350 °C. Therefore goethite could be responsible for the low blocking temperatures observed in some specimens (Fig. 5). However goethite is very widespread and occurs in specimens with blocking temperatures in excess of 400 °C, and it is not noticeably more common in specimens with low blocking temperatures (Fig. 5) so that in the majority of specimens, goethite probably does not contribute to the nrm.

The third opaque mineral phase present is the red material (ochre) that gives these rocks their characteristic colour. It stains the felspars, forms a variable proportion of the matrix, and frequently occurs as a pellicle around the silicate minerals. The ochre presumably owes its colour to finely-divided haematite.

In order to investigate the relative contributions of these phases to the stable remanence, the proportion of the intensity remaining after thermal demagnetization at 500 °C (M/M_0) was plotted against the amount of specular haematite, and against two ratios of the iron mineral content (Fig. 6). The ochre contribution was measured by ranking the specimens on a 6-point scale using thin sections made from the same specimens as the polished sections used to determine the specular haematite content. Each point was then arbitrarily assigned 0·1 per cent. All the graphs show the same trend—the most stable specimens are those with most specular haematite, but, more particularly, with the highest ratio of specular haematite to the three phases combined.

4. Stability of nrm and opaque minerals of Stac Fada Member

Ore microscopy shows that the opaque grains, almost all of which are in the matrix and not in lithic grains, are haematite. The grains occasionally contain bands or sharply defined areas of ilmenite. Alteration of haematite to goethite, and of

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ilmenite to rutile is common. Films of haematite occur, but they are volumetrically subordinate to the matrix grains. Magnetite was not identified.

The samples from this volcanic mudflow, although they have the same remanence direction, differ from other rocks of the Stoer Group in having lower blocking temperatures (500 to 600 °C, Fig. 5) and much lower coercive forces (Fig. 4). The inference is that the remanence is due to fine-grained magnetite or to ilmeno-haematite. The failure to detect magnetite optically might be taken as a reason for supposing ilmeno-haematite to be the carrier of the remanence, but very fine-grained magnetite beyond the resolving power of the microscope could be responsible.

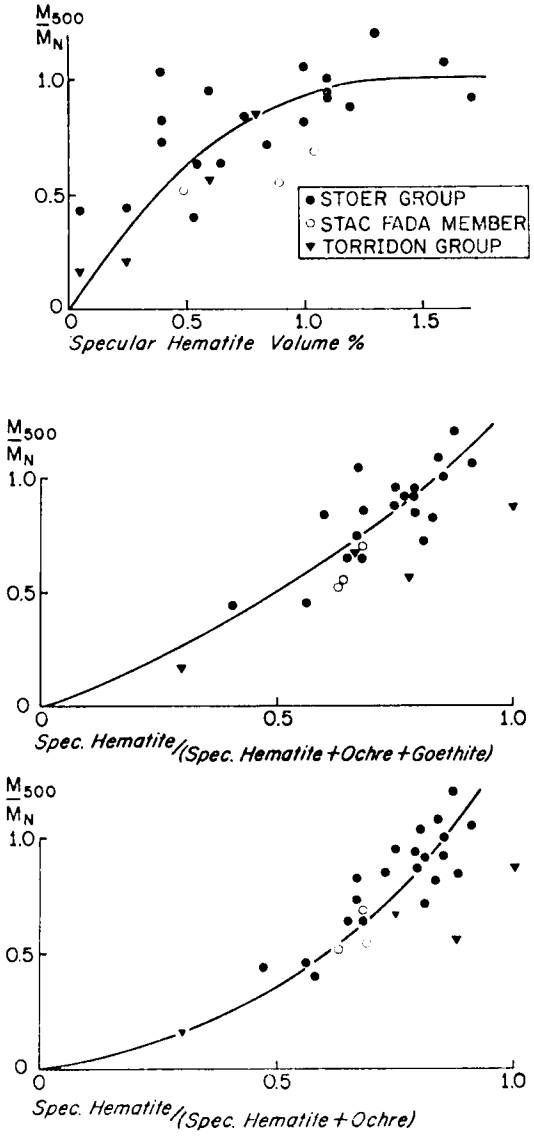


FIG. 6. Thermal stability (M_{500}/M_n) versus volume percentage of specular haematite (top), versus ratio of specular haematite to all iron minerals (middle), and versus ratio of specular haematite to all iron minerals except goethite (bottom).

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5. Field tests of stability

At Stoer Bay the Stac Fada Member can be seen to inject and disrupt the sandstone strata over which it flowed. The nrm directions in ten of the disrupted and detached blocks have previously been shown to be random ($R = 2{\cdot}50$ random at $P = 0{\cdot}25$) so that the greater part of the nrm predated the Stac Fada Member (Irving 1957). The direction of nrm in red sandstone boulders in the basal conglomerates of the Torridon Group (formerly thought to be Triassic) are random, showing that the magnetization of the Stoer Group predates the Torridon Group (Irving & Runcorn 1957). This experiment has been repeated, the conglomerates resampled and thermally demagnetized. The decay curves are comparable to those from Stoer Group sediments, and the directions, which are random, are little changed by cleaning at 500 °C

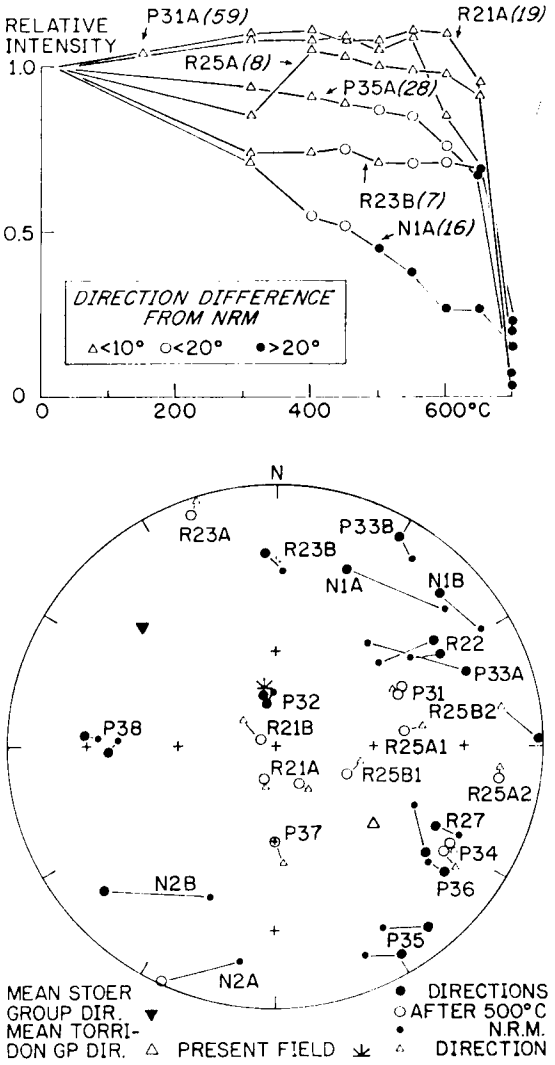


FIG. 7. Results from basal Torridon Group conglomerates. Directions of magnetization before and after heating to 500°C shown in the stereogram (primitive is horizontal plane). R samples are from Rubha Reidh, P from Achultibuie, and N from Enard Bay. For 15 cobbles and boulders (unit weight) the resultant is 5·04, no different from a random distribution at $P = 0, 05$. Normalized intensity versus temperature is plotted above. The nrm intensity (μG) is given in brackets.

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(Fig. 7). The high blocking temperatures indicate the presence of haematite, and this is confirmed by studies of polished sections, which are optically indistinguishable from Stoer sandstone *in situ*. The increase in intensity shown by specimens P31A and R21A (on thermal demagnetization) correlates with their upward direction of magnetization suggesting the presence of small components due to the present Earth's field. However these unstable components of magnetization evidently do not exceed 10 per cent of the nrm. We conclude that the nrm of Stoer sandstone is mainly due to detrital specular haematite and that it has been little affected by post-depositional events.

The magnetic stability of the Torridon Group is less easily tested. There are no intraformational conglomerates although there are intraformational slump folds. Some, but not all, of these have random directions of magnetization, suggesting a high degree of magnetic stability (Irving 1957, pp. 106–109). A fold test shows magnetization to pre-date the Caledonian deformation (Irving & Runcorn 1957, pp. 92–93). The existence of reversals with evidence of lateral continuity up to 50 km also indicates high stability, even though the directions are not exactly antiparallel (Table 3). Resistance of the magnetization to thermal cleaning in the laboratory has already been dealt with, and is indirectly confirmed by the fact that temperatures of at least 180 °C were probably attained at the base of the succession during burial in Cambro-Ordovician time without producing observable magnetic effect.

6. Analysis of directions of magnetization

The results from each sampling site are given in Tables 1 and 2. In Fig. 8 and Table 3 the locality mean directions for the Stoer Group are given, together with the earlier results. Despite the less complete sampling and absence of cleaning in the earlier study, neither the mean directions nor the dispersion are significantly changed. The directions are all of normal polarity. The Stoer Group result was obtained by averaging the 16 sites from Stoer Bay, treating Achiltibuie as 2 sites (SA1–SA8 and SA9–SA16) and Enard Bay and Rubha Reidh each as an additional site, making 20 sites in all. This is justified on the basis of stratigraphic coverage (now about 30 m per 'site' corresponding to 10^5–10^6 yr) and the possibility of initial dip at Enard Bay and Achiltibuie.

The basal Torridon Group directions are towards the south-east, and have reversed polarity. These data were mainly collected in order to map across the unconformity, and no attempt was made to penetrate deeply into the Torridon Group; the results are therefore less extensive than those from the Stoer Group. No internally consistent data was obtained from sites T_1E1, T_2E3, T_2E4, T_1A1, T_1R1, T_1R2, mostly grey sandstones. The mean direction at the other sites is essentially the same as the reversed (nrm) direction previously obtained (Fig. 8; Table 3). Thermal cleaning reduces the inclination, moving the direction away from the present field and closer to that from the normally magnetized Torridon Group. This suggests that secondary components due to the present field are responsible for the lack of exact antiparallel alignment of the normal and reversed directions, and predicts that systematic cleaning of the Torridon Group collection would bring them into alignment. This formidable task has not yet been undertaken.

Evidence of reversal periodicity is summarized in Table 4, which shows that there were order of magnitude fluctuations in the Proterozoic as in the Phanerozoic.

7. Comparisons with poles from Laurentia

The Lewisian platform used to be part of the Laurentian Shield (Du Toit 1937; Wynne-Edwards & Hassan 1970) and it is of interest to compare the above results with data from the Precambrian of North America.

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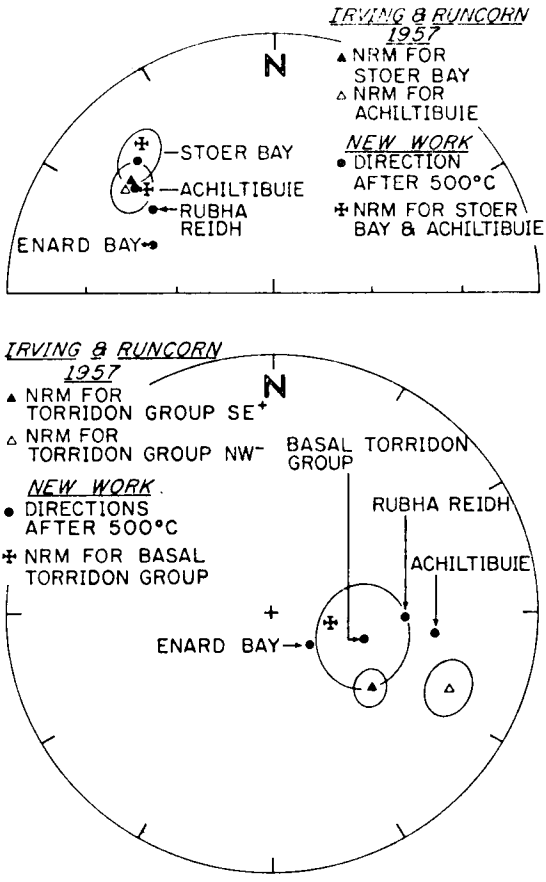


FIG. 8. Locality and overall mean directions for the Stoer Group (above) and basal Torridon Group (below). Fisher circles of confidence ($P = 0{\cdot}05$) are given. Comparisons are made between results before and after thermal cleaning, and with previous observations. Primitive is the bedding plane. All directions are plotted on the lower hemisphere. In the lower plot the directions are plotted irrespective of sign.

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Table 1

Stoer group results

Site	Nat. Grid Ref.	*M*	*N*(*R*)	*D*, *I*
Stoer Bay	58.2° N, 5.3° W			
SS1	NC042272	6·4	4(3·97)	340, +13
SS2	NC042276	4·8	4(3·81)	306, +16
SS3	NC041283	1·6	5(2·21)	random
SS4	NC038283	3·7	4(3·87)	346, −10
SS5	NC037284	16·1	4(3·96)	292, +25
SS6	NC037284	5·7	4(3·74)	316, +17
SS7	NC036284	10·6	4(3·95)	321, +36
SS8	NC035285	30·1	4(3·96)	297, +49
SS9	NC035285	10·0	4(3·88)	316, +26
SS10	NC033285	8·7	4(3·71)	310, +30
SS11	NC033286	13·1	3(2·90)	308, +37
SS12	NC033286	28·6	4(3·98)	321, +33
SS13	NC032286	12·3	4(3·99)	306, +39
SS14	NC030286	6·7	3(2·98)	315, +16
SS15	NC029286	10·3	4(1·99)	random
SS16	NC028286	14·2	4(3·97)	336, +23
SSZ	NC027288	3·7	4(3·46)	301, +40
SS17	NC027290	7·6	4(3·81)	300, +20
SS18	NC028291	7·6	4(2·87)	random
Stoer Bay mean 500° C		11·4	16(15·11)	315, +27
Stoer Bay mean NRM		13·8	16(14·88)	319, +24
Achiltibuie	58.0° N, 5.3° W			
SA1	NC025067	7·7	6(5·72)	315, +38
SA2	NC025067	7·3	6(5·24)	311, +19
SA3	NC024067	12·5	4(3·94)	301, +39
SA4	NC024067	13·7	4(3·95)	298, +30
SA5	NC024067	16·5	4(3·87)	288, +44
SA6	NC023067	25·7	4(3·92)	315, +30
SA7	NC023067	20·2	3(2·87)	315, +44
SA8	NC023067	18·7	5(4·82)	300, +32
SA9	NC023067	26·6	4(3·95)	314, +38
SA10	NC022067	37·4	4(3·96)	320, +37
SA11	NC021067	20·5	2(2·00)	286, +43
SA12	NC021067	12·5	4(3·91)	305, +33
SA13	NC020067	15·2	2(1·98)	305, +24
SA14	NC020067	14·7	4(3·96)	310, −04
SA15	NC020067	45·7	4(3·98)	305, +22
SA16	NC020067	8·9	4(3·49)	320, +51
Achiltibuie mean 500° C		19·0	16(15·47)	307, +33
Achiltibuie mean NRM		24·0	16(15·51)	310, +37
Enard Bay	58.1° N, 5.3° W			
SE1	NC033146	11·4	2(1·82)	random
SE2	NC032145	64·9	12(11·94)	283, +53
SE3	NC031145	20·2	6(5·97)	299, +44
SE4	NC025147	3·4	4(2·37)	random
Enard Bay mean 500° C		42·5	2(1·98)	292, +49
Enard Bay mean NRM		63·8	2(1·99)	291, +57
Rubha Reidh	57.9° N, 5.8° W			
SR1	NG743920	6·8	6(3·92)	354, +56
SR2	NG740919	5·5	12(11·10)	280, +41
SR3	NG746921	3·9	4(3·38)	298, +17
Rubha Reidh mean 500° C		5·4	3(2·67)	305, +42
Rubha Reidh mean NRM		6·6	3(2·71)	302, +50

M is the mean intensity of nrm (μG), *N* is the number of cores from sites or sites from localities, *R* is the resultant, *D*, *I* are the declination and inclination of the mean direction after cleaning at 500° C.

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Table 2

Results from basal Torridon group

Site	Nat. Grid Ref.	*M*	*N*(*R*)	*D*, *I*
Enard Bay	58.1° N, 5.3° W			
T_1E2	NC033147	3·4	4(3·88)	081, +74
T_1E3	NC033147	3·2	4(2·87)	random
T_1E4	NC033147	3·4	4(3·47)	159, +66
T_1E5	NC033147	2·3	4(3·98)	068, +81
T_1E6	NC033147	2·1	4(3·77)	068, +75
T_2E1	NC033147	6·3	4(3·77)	157, +53
T_2E2	NC033147	2·9	4(3·90)	160, +72
Enard Bay mean 500° C		3·4	6(5·79)	132, +75
Enard Bay mean NRM		4·2	6(5·85)	106, +83
Achiltibuie	58.0° N, 5.3° W			
T_1A2	NC025067	5·8	7(6·36)	124, +54
T_1A3	NC026067	2·3	8(6·00)	106, +40
T_1A4	NC026067	9·0	3(2·99)	075, +13
T_1A5	NC037057	0·8	8(4·13)	random
T_1A6	NC037056	0·5	5(3·33)	random
Achiltibuie mean 500° C		5·7	3(2·75)	097, +38
Achiltibuie mean NRM		8·5	3(2·77)	095, +45
Rubha Reidh	57·9°N, 5.8° W			
T_1R3	NG748919	18·6	7(6·14)	094, +36
T_1R4	NG748919	37·2	9(8·33)	087, +62
Rubha Reidh mean 500° C		27·9	2(1·95)	091, +49
Rubha Reidh mean NRM		29·2	2(1·97)	096, +56

Symbols as in Table 1.

All palaeomagnetic poles dating from the Cambrian to the Hudsonian orogenic event (1800 My) and satisfying the minimum criteria of Table 5 are plotted in Fig. 9. The results are not of equal value and in an attempt to take account of this variability, each pole has been graded on a seven-point reliability scale. The criteria are set out in Table 5 and the numerical details in Table 6. Each pole has a symbol whose radius is proportional to trade. The aim of the map is to summarize data on the location of the pole at specified times. Data purposely excluded are from the Sudbury complex, for which the structural interpretation is obscure, and the Keweenawan lavas at Alona Bay and Cape Gargantua which immediately overlie pre-Keweenawan basement and which therefore may have had initial dips. Results failing to reach certain minimum criteria (the ' B ' data of Hicken *et al.* 1972), such as those from the Baragar County, Marathon, and Molson dykes, are also excluded. New age assignments for the Mugford volcanics at 1150[d] (Sutton *et al.* 1972) and the ' lowermost ' Keweenawan lavas at 1200[b] (now regarded as Lower Keweenawan by Hubbard 1972, p. 627) are included.

Ideally, each age datum should be ascribed an error deduced from a consideration of the analytical precision, the chemical suitability of the analysed material, and the geological significance of the calculated age. This was beyond our expertise. Instead, all age data have been slotted into three arbitrary classes intended to provide 95 per cent confidence limits for the timing of the last ' major ' thermal event. Generally, the more work done the higher the assigned reliability. Needless to say, the procedure cannot do justice to all data. Some Rb–Sr isochrons from rocks with a wide range of Rb/Sr ratios, high analytical precision, and a uniform initial Sr^{87}/Sr^{86} ratio, should

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Table 3

Mean directions and poles

Rock Unit	Polarity	Treatment	*D*, *I*	*N*(*R*)	*k*	α_{95}	Pole (*dm*, *dp*)
Torridon Group (Irving & Runcorn 1957)	N	NRM	294, −28	28(25·14)	9	9	00 , 236E(10, 06)
Torridon Group (Irving & Runcorn 1957)	R	NRM	129, +51	53(49·45)	15	5	09S, 217E(07, 05)
Basal Torridon Group (this work)	R	NRM	099, +73	11(10·42)	17	11	43S, 219E(20, 18)
Basal Torridon Group (this work)	R	500° C	106, +61	11(09·95)	10	16	28S, 228E(24, 19)
Stoer Group (this work)	N	NRM	316, +28	20(18·66)	14	9	36N, 230E(10, 05)
Stoer Group (this work)	N	500° C	313, +29	20(18·95)	18	8	35N, 234E(09, 05)
Stoer Group (Irving & Runcorn 1957)	N	NRM	307, + 34	13(12·70)	40	7	35N, 248E(08, 05)

Symbols as for Table 1 with these additions: k is estimate of precision, α_{95} the circle of confidence $p = 0{\cdot}05$ (Fisher 1953), *dm* and *dp* are the polar errors perpendicular to and in the direction of the meridian through the sampling locality (Creer *et al.* 1957).

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Table 4

Lengths of polarity intervals (My)

Group	t_1	t_2	n	T
Torridon	10–20	30–150	>17	~1·5
Stoer	—	11–60	0	>11

t_1 is the duration of sedimentation estimated from the isotopic results (Moorbath 1969, p. 161); t_2 is the duration estimated by comparison with Phanerozoic rocks of similar lithofacies; n is the number of reversals (Irving & Runcorn 1957, p. 90); T is the average length of polarity intervals. T for the Torridon has been calculated using 25 My as the best current estimate of the duration.

perhaps, be rated three points instead of two. Criticism might also be directed at the lumping of whole-rock and mineral ages, since these may record different thermal events. Frequently, however, the mineral separations described in the literature are mixtures, such as biotite and hornblende, with no grain sizes specified, and we therefore adopt the conservative procedure of giving them equal weight. In grading the palaeomagnetic data however, we feel ourselves on firmer ground, although many features of the criteria used could be criticized.

These criteria may at first sight appear stringent. However, poles allotted all available points would have $\alpha_{95} \simeq 15°$ if the age uncertainty is expressed in angular terms using the factor 0·3° of apparent polar wander per million years (Donaldson *et al.* 1973). No pole reaches this standard. At the other extreme, when the age uncertainty reaches ± 200 My, the equivalent polar uncertainty expands to 60°, even for data with very high palaeomagnetic precision—a feature of Precambrian work that has already been commented upon by Spall (1971) and Brock (1973).

In Fig. 10 an attempt is made to construct a path of apparent polar wandering using the data of Fig. 9 as a basis. The pole should have passed within 10° of any of those poles allocated two or more points for palaeomagnetic accuracy, even if not at the times assigned. Further evidence (summarized beneath Table 6) to be taken into account are the superimposition and cross-cutting relationships of rock units. The simplest path that can be constructed, and ignoring for the moment the possibility of regional rotations, is shown in the upper part of Fig. 10. This path differs little from that given by Irving & Park (1972, Fig. 2) and is indeed based on virtually the same data. The polar loop between 1200 and 1000 My BP, is essentially the same as that given by Du Bois (1962), Black (1963), and Robertson & Fahrig (1971), and implied in Palmer (1970). A notable departure from the reconstruction of Irving & Park (1972) however, is the tight loop south of the equator which encompasses the Grenville poles (817, 950[d] and 1000[b]). The early part of such a loop is already to be seen in the reconstructions of Du Bois (1962) and Black (1963).

The Stoer and Torridon poles, after rotation for closure of the Atlantic, fall in the *general* area of late-Precambrian poles for Laurentia. The poles for the transcaledonian Rogaland basement complex of Norway, and the hyperite-dolerite dykes of Sweden, are some distance to the east. There are however, anomalies. The Torridon poles lie near the edge of the poles for the much older Mackenzie igneous episode (1200 My). The Stoer pole is less obviously misplaced near the slightly older poles, mainly Keweenawan, dated at around 1100 My BP. A better fit is achieved by rotating north-west Scotland 12° clockwise about a local axis. This brings the mean Torridon Group pole within 14° of the Franklin (675 My BP) and the Freda–Nonesuch poles (1020 My BP) and within the limits of the possible path of Fig. 10. There is no other evidence for such rotation, but it is a possibility in areas such as the margins of

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Table 5

Grading of Precambrian[1] Palaeomagnetic Poles

Criterion	Standard	Points	Points
I Accuracy[2] of palaeomagnetic data	$k_0N \geqslant 625$	3	k_0 is Fisher's estimate of precision (Fisher 1953) normalized to equator using Model A[3] to allow comparison between results from different latitudes. N is the number of sites or localities used in computing k_0. Localities with <10 samples or <4 sites are not admitted.
	$625 > k_0N \geqslant 225$	2	
	$225 > k_0N$	1	
II Accuracy of rock age[1]	± 50 My	3	Based on two or more isotopic methods (>9 determinations if K–Ar[4]) giving concordant results.
	± 100 My	2	Based on one or more isotopic methods (>9 determinations if K–Ar[4]) giving concordant results within the range specified.
	± 200 My	1	May be based on <10 K–Ar determinations, or poorly concordant results from different isotopic methods, or on circumstantial evidence (e.g. age of similar rocks in adjacent areas known).
III Time of acquisition of palaeomagnetism[5]	= Rock age	3	Intraformational conglomerate has random directions. Fully executed contact test in igneous rocks.
	Old	2	AF or thermal cleaning *or* areally extensive reversals in sediments/lavas *or* fold/conglomerate test.
	Uncertain	1	Natural remanence only.

Notes to Table 5

(1) Phanerozoic rock ages are several times more accurate than those in the Precambrian and different grades will be needed for the Phanerozoic—say ± 10 ± 20 ± 40 My. For uniformity Cambrian results are graded by the Precambrian scale in Table 6 which almost certainly rates them too highly.

(2) The radius of the circle of confidence $P = 0{\cdot}05$ around the mean direction is given approximately by $140(kN)^{-\frac{1}{2}}$, so that, as a rule-of-the-thumb, three points correspond to a direction error (reduced to the equator) of $<6°$, two points 6–13°, and one point $>13°$.

(3) Model A (Irving & Ward 1964), modified to a small degree that is negligible for these purposes, fits both the recent (Cox 1970) and ancient (Brock 1972) Earth's field within the accuracy of present observations.

(4) Maximum K–Ar age is used in Table 6.

(5) It is difficult, sometimes impracticable, to show that the magnetization of a rock was acquired at the time it was formed. Most tests (e.g. fold test (Graham 1949)) are capable only of giving a minimum age of magnetization—limits which may represent only a small proportion of the rock age. Although, in particular instances, it may be safe to assume that the successful completion of such tests implies contemporaneity, such extrapolations cannot confidently be made in a general way. For example the presence of reversals may be taken to imply contemporaneity in Tertiary basalts, but this may not always be justified because reversals occur in high-grade metamorphic rocks (Balsley & Buddington 1957; Hargraves & Burt 1968; Irving *et al.* 1972a) which certainly have a secondary magnetization. For these reasons the majority of palaeomagnetic tests are allotted two points, the highest rating being reserved for those results for which contemporaneity can at this time be established beyond reasonable doubt from the study of penecontemporaneous conglomerates or a fully executed contact test (Bruhnes 1906; Everitt & Clegg 1962).

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Table 6

Index of palaeomagnetic poles

Age (My)	Rock unit	Place	I	II	III	Grade	Ref.
500	Lamotte Formation IIA	Missouri	1	3	2	6	2–076
525	Wichita Granite	Oklahoma	1	1	2	4	2–045
540a	Wilberns Formation	Texas	1	3	1	5	2–019
540b	Lodore Formation	Utah	2	3	1	6	2–021
675	Franklin diabase	Northern Canada	3	2	2	7	1–313
700	Grenville dykes	Ontario, Quebec	1	1	2	4	1–161
796a	Torridon Group (normal	NW Scotland	1	2	3	6	1–1
796b	Torridon Group (reversed)	NW Scotland	3	2	2	7	1–1
817	Frontenac dykes	Ontario	2	1	2	5	(1)
950a	Grenville basic rocks	Ontario	1	1	2	4	(2)
950b	Rogaland basement	SW Norway	1	2	2	5	(3)
953	El Paso igneous rocks	Texas	3	2	3	8	1–167
991	Stoer Group	NW Scotland	2	2	3	7	herein
1000a	Hyperite dolerite	Southern Sweden	1	1	2	4	1
1000b	Allard Lake anorthosite	Quebec	1	1	2	4	1–130
1020	Freda and Nonesuch sediments	Lake Superior	2	2	2	6	1–303
1050a	Logan diabase (normal)	Lake Superior	2	2	2	6	1–268
1050b	Logan diabase (reversed)	Lake Superior	3	2	3	8	1–309
1050c	Aillik dykes	Labrador	1	1	2	4	(8)
1076a	Mamainse lavas (normal)	Lake Superior	3	2	2	7	1–141
1076b	Mamainse lavas (reversed)	Lake Superior	1	2	2	5	1–140
1080	Copper Harbour Group	Lake Superior	1	1	2	4	1–132
1100a	Mellen gabbro	Wisconsin	1	1	2	4	1–2161
1100b	Osler volcanics	Lake Superior	1	1	2	4	1–147
1100d	Portage Lake Lavas	Lake Superior	2	1	2	5	1–305
1115a	Keweenawan intrusives (normal)	Minnesota	3	3	2	8	1–187
1115b	North Shore volcanics (normal)	Lake Superior	3	2	2	7	1–146
1115c	North Shore volcanics (reversed)	Lake Superior	3	2	2	7	1–306
1130	Mackenzie dykes	Canada	3	2	3	8	1–116, 117(4)
1150a	Gila County diabase	Arizona	2	2	2	6	1–320
1150b	Abitibi dykes	Ontario, Quebec	2	1	2	5	1–109
1150c	Muskox intrusion	NW Territories	2	1	2	5	1–115
1150d	Mugford volcanics	Labrador	2	1	2	5	1–314(5)
1200a	Coppermine lavas	NW Territories	2	2	2	6	1–121
1200b	Ironwood lavas	Lake Superior	2	1	2	5	1–150
1200c	Michipicoten volcanics	Lake Superior	2	1	2	5	1–142
1200d	Gargantua volcanics	Lake Superior	2	1	2	5	1–143
1225	Sudbury dykes	Ontario	2	2	3	7	1–119, 120
1350a	St. Francois igneous rocks	Missouri	3	2	2	6	1–240
1350b	Beltian rocks	Alberta, Montana	2	1	1	4	1–225
1370	Sibley Formation	Lake Superior	1	2	2	5	(6)
1410	Sherman Granite	Colorado, Wyoming	2	2	2	6	1–172
1475	Croker Island Complex	Ontario	1	2	2	5	1–129
1500a	Michikamau anorthosite	Labrador	1	1	2	4	1–205
1500b	Michael Gabbro	Labrador	3	2	2	7	(8)
1600a	Gunflint Formation	Ontario	1	2	2	5	1–191
1600b	Western Channel diabase	NW Territories	2	1	3	6	(6)
1600c	Et-Then Group	NW Territories	2	1	2	5	(4)
1830	Dubawnt Group	NW Territories	2	2	2	6	(7)

Notes to Table 6

Results are indexed by the best estimate of the rock age obtained from the literature. Rb–Sr ages have been recalculated using decay constant of $1{\cdot}39 \times 10^{-11}$ yr^{-1}. These ages are subject to the uncertainties indicated in column II. The Mugford volcanics have been assigned an age conformable with the correlation of Sutton *et al.* (1972). The Rb–Sr studies of Gates (1971) are conformable with the age assignment for the Sudbury dykes given here, but yield an age (2100My) much older than that given here for the Abitibi dykes. The results are graded by the criteria of Table 5; Column I gives the accuracy of the palaeomagnetic data, Column II the accuracy of the rock age, and Column

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III the 'contemporaneity' rating, which added together give the grade. For reasons given in the footnote to Table 5 the Cambrian data is probably rated too highly. Reference to locality and pole co-ordinates and to originals is made by the catalogue number of Hicken *et al.* (1972). Late results not yet entered in that catalogue are as follows: (1) Park & Irving (1972); (2) Irving *et al.* (1972a); (3) Poorter (1972); (4) Irving *et al.* (1972b); (5) Murthy & Deutsch (1972); Robertson (1973); (6) Irving *et al.* (1972c); (7) Park *et al.* (1973); (8) Fahrig & Larochelle (1972).

When constructing polar paths account must be taken of geological relationships as well as radiometric ages. In so far as they are known to the authors these are as follows: 1050b intrudes 1370; 1050a intrudes 1050b; 1115b overlies 1115c; 1020 overlies 1080 which overlies 1100d; 1130 intrudes 1600c.

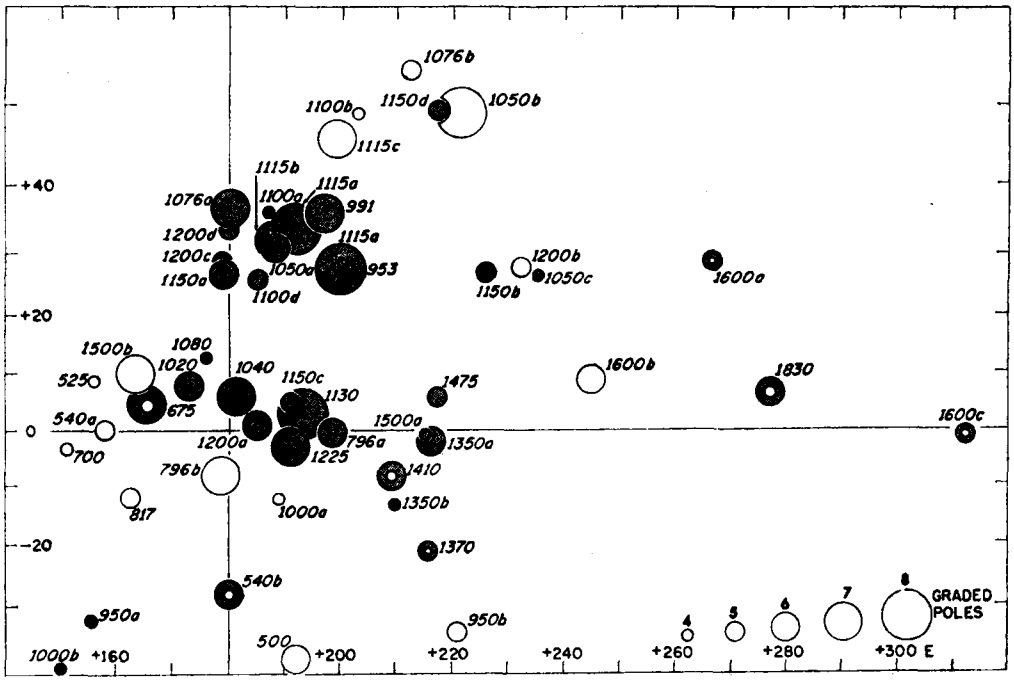


FIG. 9. Graded palaeomagnetics poles for Laurentia from the Cambrian to the end of the Hudsonian event. Mercator projection. Normal poles (defined by north-seeking magnetization vectors directed towards the Pacific) are solid. Reversed poles are open circles. Mixed polarity poles with statistically antiparallel magnetization vectors are shown as bull's eyes. Poles are labelled with their assigned ages (Table 6). Symbol size is derived from the weighting criteria (Table 5). Poles from Scotland and Norway are rotated for closure of North Atlantic as in Fig. 10.

Laurentia that have undergone repeated fracture. The poles from El Paso (953) and the Wilberns Formation (540^b) that are also anomalous may also reflect marginal rotations about local axes.

Another possible reconstruction (Fig. 10 alternative 1) utilizes the idea (Palmer & Carmichael 1972; Irving *et al.* 1972a) that the Grenville Plate has moved relative to the rest of Laurentia (interior Laurentia). The Grenville Plate is that part of the Grenville Province south of a hypothetical suture situated a distance of the order 100 km south of the Grenville Front (Donaldson & Irving 1972). The path for interior Laurentia is the same as that shown in Fig. 10 for the period prior to 1000 My BP, at which time the pole halts until joined by the northward moving 'Grenville' pole. After about 700 My BP the two then move westwards together.

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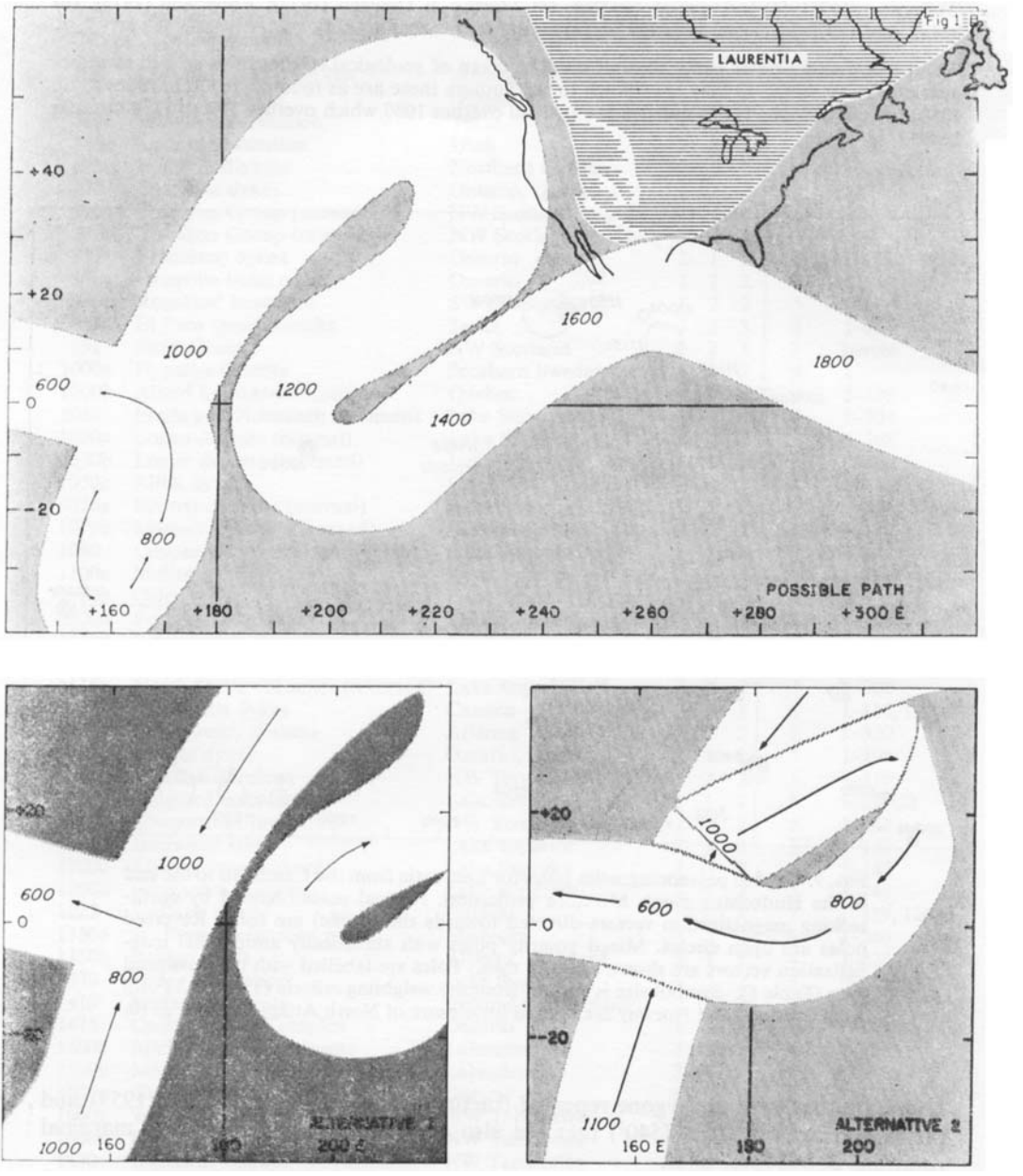


FIG. 10. Optional polar paths for Laurentia. The approximate edges of Laurentia are shown by stripes. Greenland has been rotated relative to North America in accordance with the reconstruction of Bullard *et al.* (1965). The sampling area for the Stoer and Torridon Groups is shown in the top right-hand corner.

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A third reconstruction, assuming displacement of the Grenville Plate but this time without assuming the rotation of margins of Laurentia, is made in Fig. 10 (alternative 2). The pole does not halt at 1000 My, but between then and 700 My executes a subsidiary loop encompassing the Stoer (991), El Paso (953), Torridon (796^a 796^b), and Frontenac dyke (817) poles.

The possible path and alternatives 1 and 2 of Fig. 10 correspond to three different interpretations of the late Proterozoic tectonic history of Laurentia. The possible path is consistent with the usual opinion that the Grenville Province has developed in close proximity to interior Laurentia, as for example implied by Wynne-Edwards (1972, p. 326); indeed the path is constructed on the assumed integrity of Laurentia since 1800 My BP. On the other hand the alternative reconstructions of Fig. 10 assume that the Grenville Plate (see above) has moved by several thousand kilometres relative to interior Laurentia. The times at which polar paths merge mark the times at which plates, previously moving separately, began moving together; they are the polar signatures of the collision and suturing of continent-bearing lithosphere. Hence the polar junctures of alternatives 1 and 2 may correspond to the suturing of the Grenville Plate to interior Laurentia. Alternatives 1 and 2 differ in the age assigned to the magnetization of the high-grade metamorphic Grenvillian basement. Under alternative 1 the remanence is assigned (Du Bois 1962; Hargraves & Burt 1968) to the conspicuous thermal event (corresponding to temperatures ~200 °C, Wynne-Edwards 1972, p. 264) recorded by the K–Ar ages at about 1000 My BP (Stockwell 1970), so that the polar juncture, and the collision it might imply post-dates the K–Ar thermal event by roughly 10^8 yr. Under alternative 2 the polar juncture coincides approximately in time with the K–Ar thermal event and the deposition of the Stoer Group, both of which may therefore have been caused by uplift following suturing of the Grenville Plate to interior Laurentia. The age of magnetization is assigned following Irving *et al.* (1972a) to 1100 My BP when the effect of regional metamorphism, dated at about 1200 My BP by Wynne-Edwards (1972, p. 264) was waning. That is, the remanence is supposed to 'see through' the later K–Ar event. Clearly, however, on the basis of the existing data, graded according to the criteria of Table 5, it is not possible to decide between these alternatives. Nor does Fig. 10 exhaust the possibilities.

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